Program : **Diploma in Printing Technology**	
Course Code : **3104**	Course Title: **Fundamentals of Color Reproduction**
Semester : **3**	Credits: **4**
Course Category: **Program Core**	
Periods per week: **4 (L:3, T:1, P:0)**	Periods per semester: **60**

Course Objectives:

- To guide the student to the basics of color and to learn how it influences the real printing world.
- To provide exposure to the fundamentals of color reproduction.
- To impart color concept to the printing process and to develop idea on most exciting possibilities of color reproduction in the upcoming era.

Course Prerequisites:

Topic	Course code	Course Title	Semester
Basics of color and light		Applied Physics I	I

Course Outcomes:

On completion of the course student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Explain the basics of color and its role in image reproduction.	15	Understanding
CO2	Describe the role of CIE in color reproduction and the basic aspects of color models and color gamut.	14	Understanding
CO3	Describe the basics and background of color measurement and color management process.	15	Understanding
CO 4	Assess print quality based on optical density and color values.	14	Applying
	Series Test	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	2						
CO3	2						
CO4	3	3					

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Explain the basics of color and its role in image reproduction.**		
M1.01	Introduction to Color.	1	Understanding
M1.02	Color and light.	1	Understanding
M1.03	Human Color perception.	2	Understanding
M1.04	Color Schemes.	2	Understanding
M1.05	Maxwell Color Triangle & Color theories.	3	Understanding
M1.06	Color reproduction in Printing.	2	Understanding
M1.07	Color Separation.	2	Understanding
M1.08	Basics of Scanners.	2	Understanding
Contents: Color: Definition. Attributes of color - Hue, Saturation, Lightness, Alliance of EM Spectrum, Wavelength of Light and Color, Spectral power distribution curve. Human Color perception-Elements of eye, Rod and LMS Cone cell vision. Color Schemes: Warm color, Cool color, Monochromatic, Analog and Complementary color. Color wheel. Memory color, Metamerism. Color Theories: Additive and Subtractive color theories. Color theories Vs Color perception and Color reproduction. Maxwell Color Triangle. Color reproduction: Elementary principles of Multi color printing. Color Separation: Conventional and Electronic principles of color separation. Scanners - Working Principle & Types.			
CO2	**Describe the role of CIE in color reproduction and the basic aspects of color models and color gamut.**		
M2.01	CIE definition and milestones.	2	Understanding
M2.02	CIE color matching experiment 1931.	2	Understanding
M2.03	Standard Observer.	1	Understanding
M2.04	Tristimulus values.	2	Understanding
M2.05	Color matching functions.	2	Understanding
M2.06	Chromaticity diagram.	2	Understanding
M2.07	Color Gamut.	2	Understanding

M2.08	Color Models or Color spaces.	1	Understanding
	Series Test I	1	
Contents: Commission Internationale de l´Eclairage(CIE) - definition. CIE milestones, Color matching experiment 1931.Standard Observer, Tristimulus values & Color Matching functions. Chromaticity coordinates and Chromaticity Diagram. Standard illuminant - Definition, Types of illuminants, Color temperature. Color Gamut - Definition, Color gamut of different devices used in Print production (Scanner, Digital camera, Monitor, Proofing devices, Press). Color Models: Muncell, RGB, CMY, HSB, CIE Color spaces - CIE XYZ, CIE L*a*b*, CIE LUV. Device dependant and independent color Models.			
CO3	**Describe the basics and background of color measurement and color management process.**		
M3.01	Color measurement significance and basic elements.	3	Understanding
M3.02	Basic working principle of color measurement.	3	Understanding
M3.03	Digital workflow and importance of accurate color transformation in image reproduction process.	3	Understanding
M3.04	Basic aspects of color management.	3	Understanding
M3.05	Components of Color Management System.	3	Understanding
Contents: Color measurement- Significance, elements required to see the color and measure the color. Tristimulus color measurement, Color measuring devices-Colorimeter, Spectrophotometer-components and working principles. Factors affecting measurement of color. Color management - Definition and importance. Digital Workflow system and Color management, WISYWYG Principle, Open loop and Closed loop color management, 3C's of Color Management - Calibration, Characterization, and Conversion. ICC profile in Color Management. Components of CMS- Device profile, PCS, CMM and Rendering indents.			
CO4	**Assess print quality based on optical density and color values.**		
M4.01	Halftone image	3	Understanding
M4.02	Optical Density & Tonal Gradation	4	Understanding
M4.03	Densitometry, Colorimetry and Spectrophotometry-concept.	4	Understanding
M4.04	To apply knowledge on tone and color assessment of a print.	3	Applying
	Series Test II	1	
Contents: Halftone image - Significance and basics of halftone imaging. Halftone dots - features and types. Optical Density - Definition, play role in printing. Tonal Gradation - Definition. Method of evaluating tonal gradation of process color print. Densitometry, Colorimetry and Spectrophotometry-concept. Differentiate Density and Color. Print quality parameter - Definition and Formula to evaluate - Print contrast, Hue error, Grayness, Ink Trapping, Dot gain & Delta E. Dot area measurement formula by Murray Davis and Yule Neilson.			

Text / Reference:

T/R	Book Title/Author
T1	P. Green, **Understanding Digital Colour,** GATF, 1995
R1	A. Sharma, **Understanding Colour Management** ,Thompson Dalmar Learning
R2	**The Reproduction of Color,** The Wiley –IS&T Series in Imaging Science and Technology.
R3	Miles South worth-**Color Separation Techniques.**
R4	Yule, John A.C., **Principles of Color Reproduction.**
R5	Morovic, J., **Color Gamut Mapping.**

Online resources:

Sl. No	Website Link
1	https://en.wikipedia.org/wiki/Color
2	https://last.hit.bme.hu/download/firtha/video/Colorimetry/Hunt_Reproduction_of_Color.pdf
3	http://printwiki.org/Color